

Logic and Proof

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Introduction

A *proposition* is a declarative sentence with a definite truth value in $\{\text{True}, \text{False}\}$. A *proof* is a finite chain of propositions, each justified by an axiom or a prior theorem under valid rules of inference. Every theoretical claim in mathematics — and every formal guarantee in a machine-learning paper — ultimately reduces to such a chain. The goal of this lesson is to make the vocabulary precise and to drill the four basic proof strategies.

First, an example

Before any formal apparatus, look at one truth table and one short proof.

Truth table for $p \Rightarrow q$. The implication “if p then q ” is false only in the single case where the hypothesis p is true and the conclusion q is false:

p	q	$p \Rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

Three-step direct proof: the sum of two even integers is even.

1. Let $m = 2a$ and $n = 2b$ for some integers a, b (definition of even).
2. Then $m + n = 2a + 2b = 2(a + b)$ (factor).
3. Since $a + b$ is an integer, $m + n$ is twice an integer, hence even.

That is the entire skeleton of a direct proof: unfold the hypothesis, do algebra, recognise the conclusion. The rest of this lesson formalises the vocabulary and adds three more strategies.

Definitions

Definition 1 (Proposition). *A proposition is a statement that is either true or false (but not both). Variables p, q, r range over propositions.*

Definition 2 (Logical connectives). *For propositions p, q we define:*

$$\begin{aligned}\neg p &: \text{“not } p\text{”, true iff } p \text{ is false;} \\ p \wedge q &: \text{“} p \text{ and } q\text{”, true iff both are true;} \\ p \vee q &: \text{“} p \text{ or } q\text{”, true iff at least one is true;} \\ p \Rightarrow q &: \text{“if } p \text{ then } q\text{”, false iff } p \text{ true and } q \text{ false;} \\ p \Leftrightarrow q &: \text{“} p \text{ iff } q\text{”, true iff } p \text{ and } q \text{ have the same value.}\end{aligned}$$

Read aloud: “not p ” is true exactly when p is false; “ p and q ” needs both true; “ p or q ” needs at least one true; “ p implies q ” fails only when p is true and q is false; “ p iff q ” holds when p and q share the same truth value.

Definition 3 (Tautology). *A compound proposition is a tautology if it is true under every assignment of truth values to its atomic variables.*

Definition 4 (Predicate, quantifiers). *A predicate $P(x)$ on a domain D is a statement whose truth value depends on $x \in D$. The universal quantifier $\forall x \in D, P(x)$ asserts P holds for every x ; the existential quantifier $\exists x \in D, P(x)$ asserts P holds for at least one x .*

Properties and Theorems

Proposition 1 (Contrapositive equivalence). *For all propositions p, q , $(p \Rightarrow q) \Leftrightarrow (\neg q \Rightarrow \neg p)$.*

Proposition 2 (De Morgan). *$\neg(p \wedge q) \equiv (\neg p) \vee (\neg q)$ and $\neg(p \vee q) \equiv (\neg p) \wedge (\neg q)$. For quantifiers, $\neg \forall x P(x) \equiv \exists x \neg P(x)$ and $\neg \exists x P(x) \equiv \forall x \neg P(x)$.*

Theorem 1 (Principle of Mathematical Induction). *Let $P(n)$ be a predicate on $\mathbb{N} = \{1, 2, 3, \dots\}$. Suppose*

1. *(Base case) $P(1)$ is true, and*
2. *(Inductive step) for every $k \in \mathbb{N}$, $P(k) \Rightarrow P(k+1)$.*

Then $P(n)$ is true for every $n \in \mathbb{N}$.

Worked Examples

Direct proof

Proposition 3. *If n is an even integer, then n^2 is even.*

Proof. By hypothesis $n = 2k$ for some $k \in \mathbb{Z}$. Then $n^2 = (2k)^2 = 4k^2 = 2(2k^2)$, which is twice an integer, hence even. $\square$

Contrapositive

Proposition 4. *If n^2 is an even integer, then n is even.*

Proof. We prove the contrapositive: if n is odd, then n^2 is odd. Write $n = 2k + 1$. Then $n^2 = 4k^2 + 4k + 1 = 2(2k^2 + 2k) + 1$, which is odd. $\square$

Contradiction

Proposition 5. *$\sqrt{2}$ is irrational.*

Proof. Suppose for contradiction $\sqrt{2} = a/b$ with $a, b \in \mathbb{Z}$, $b \neq 0$, $\gcd(a, b) = 1$. Then $a^2 = 2b^2$, so a^2 is even, so a is even (contrapositive lemma above). Write $a = 2c$. Then $4c^2 = 2b^2$, so $b^2 = 2c^2$, so b is even. But then $\gcd(a, b) \geq 2$, contradicting $\gcd(a, b) = 1$. $\square$

Induction

Theorem 2 (Closed form for the first n positive integers). *For every $n \geq 1$,*

$$\sum_{k=1}^n k = \frac{n(n+1)}{2}.$$

Read aloud: “the sum, as k runs from 1 to n , of k , equals n times n plus one, all over two.”

Proof. We argue by induction on n , applying Theorem 1.

Base case. For $n = 1$ the left-hand side is $\sum_{k=1}^1 k = 1$ and the right-hand side is $1 \cdot 2/2 = 1$. Both equal 1.

Inductive step. Fix $m \geq 1$ and assume the inductive hypothesis

$$\sum_{k=1}^m k = \frac{m(m+1)}{2}. \quad (\text{IH})$$

Read aloud: “the sum, as k runs from 1 to m , of k , equals m times m plus one, over two — this is the inductive hypothesis.” We must show the statement for $n = m + 1$. Compute

$$\begin{aligned}\sum_{k=1}^{m+1} k &= \left(\sum_{k=1}^m k \right) + (m+1) \\ &\stackrel{(\text{IH})}{=} \frac{m(m+1)}{2} + (m+1) \\ &= (m+1) \left(\frac{m}{2} + 1 \right) \\ &= (m+1) \cdot \frac{m+2}{2} \\ &= \frac{(m+1)((m+1)+1)}{2},\end{aligned}$$

Read aloud: “the sum from $k = 1$ to $m + 1$ equals the sum from $k = 1$ to m plus the new term $(m + 1)$; substitute the inductive hypothesis; factor out $(m + 1)$; simplify; recognise the closed form at $n = m + 1$.” which is the claim for $n = m + 1$.

By Theorem 1, the identity holds for every $n \geq 1$. □

Connection to ML

Every formal guarantee in a machine-learning paper is a quantified proposition: a convergence rate is $\forall \epsilon > 0, \exists N, \forall n \geq N, \|\theta_n - \theta^*\| < \epsilon$; a generalisation bound is $\forall \delta, \Pr[\text{risk gap} \leq f(\delta)] \geq 1 - \delta$. Identifying the proof strategy — direct construction, contrapositive, contradiction, or induction (often on network depth or training step) — is the first step in critiquing the paper. For example, a Lipschitz bound for a deep network is classically proved by induction on the layer index, with the inductive hypothesis that the partial network up to layer k has Lipschitz constant $\prod_{i \leq k} L_i$.

Exercises

1. (Direct.) Prove that the sum of two odd integers is even.
2. (Contradiction.) Prove that there is no largest prime. *Hint:* if $p_1, \dots, p_n$ were all the primes, consider $N = p_1 \cdots p_n + 1$.
3. (Induction.) Prove that $\sum_{k=1}^n k^2 = \frac{n(n+1)(2n+1)}{6}$ for every $n \geq 1$.

4. (Truth tables.) Verify by truth table that $(p \Rightarrow q) \Leftrightarrow (\neg q \Rightarrow \neg p)$ is a tautology.